

One Dialogue, Multiple Outcomes: Objective-Specific Assessment with Matrix-Aware Muon-LoRA

Team `load_state_dict`

Trace the Ace · K-12 AI Infrastructure Competition Solution Write-up

Code and audited artifacts: github.com/chuxiliyixiaosa/trace-the-ace-muon-lora

Abstract

One tutoring dialogue can support different outcomes for different learning objectives. We apply Muon to Qwen3-4B’s LoRA factors and classification matrix while freezing the backbone; runtime audits verify this assignment. With optimizer-specific learning-rate tuning and fixed robust Platt calibration, the single model reaches 0.591436 Log Loss and 0.653200 AUROC in post-competition public/A analysis, versus 0.593883 and 0.650021 for AdamW. Its Log Loss is lower than the first-place public/A leaderboard score. Paired session bootstrapping supports the Log Loss gain; AUROC is a higher point estimate. Three lessons follow: assess individual objectives, validate late-evidence retention, and audit probability quality separately from ranking. No model ensemble is required.

1. Key Findings

The task is to predict whether a student will answer the next objective-aligned item correctly from a tutoring transcript. A model must identify evidence relevant to the target objective, rather than score overall session quality [10, 9]. In a 47-minute session, the tutor praises the student for independent work, yet the student later answers the target item incorrectly. AdamW assigns a 0.659 success probability; Muon-LoRA lowers it to 0.471. Fluent participation and scaffolded success need not imply independent mastery.

Our official submission ranked fourth on private/B: its AUROC exceeded the second- and third-place systems, but its Log Loss did not. We treat ranking and calibration separately [5, 6]; Muon-LoRA improves both observed metrics over AdamW on the same public/A examples (Table 1). We observed better private/B than public/A scores for nearly all teams, including our submission. This motivates our expectation of a potentially larger Muon lead on private/B, an extrapolation rather than a measured result: Muon was not submitted for private/B evaluation. Muon was originally proposed for full-parameter matrix training [2]. We route it instead through every trainable LoRA factor and the classification matrix under a frozen backbone. This produces our strongest audited single model on the released benchmark; runtime membership records verify the optimizer path rather than relying on configuration declarations. Three lessons connect this recipe to educational assessment. **Assess the objective, not the session:** 709 sessions have different outcomes across objectives, so fluent dialogue is not a universal mastery signal. **Validate late-evidence retention:** the context audits favor preserving later dialogue, but only within a tested checkpoint-format pair. **Separate ranking from probability quality:** a useful ordering does not ensure reliable success probabilities. These lessons guide assessment design, not causal claims about teaching. XGBoost and fusion remain optional extensions.

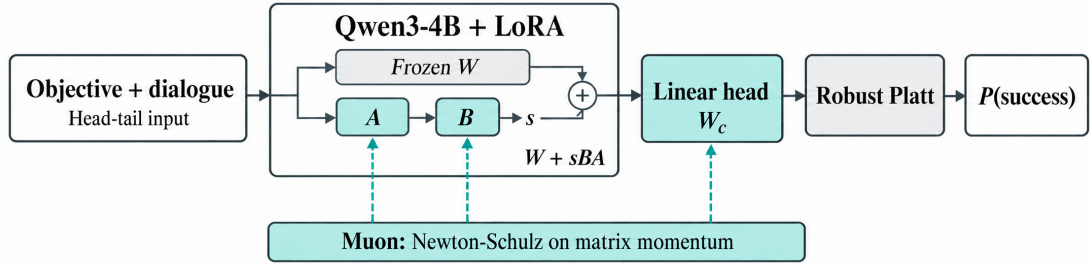


Figure 1: Single-model Muon-LoRA, showing one adapted linear module; other Transformer operations are omitted. Solid arrows: inference. Dashed arrows: training-only updates to A , B and the classifier. W stays frozen.

Table 1: Single-model results and an optional ensemble extension. Splits are reported separately; bold marks the principal single-model result.

System	Split	Log Loss	AUROC
Private/B leaderboard #1 team, oleh	private/B	0.59233	0.65088
Our official submission, rank 4	private/B	0.59449	0.64985
Our same official prediction file	public/A	0.59663	0.64465
AdamW + robust Platt	public/A	0.593883	0.650021
Muon-LoRA 2.5e-5 + robust Platt	public/A	0.591436	0.653200
<i>Optional ensemble extension</i>			
0.4 AdamW XGB-28 β + 0.6 Muon 2.5e-5 head β	public/A	0.590136	0.657522

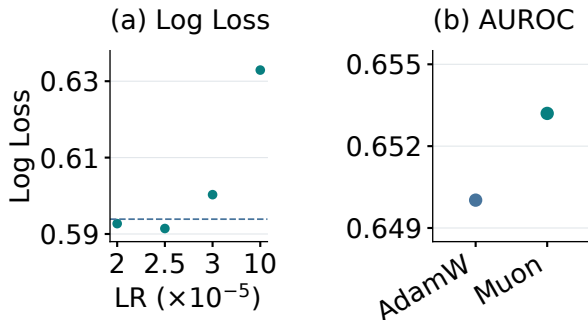


Figure 2: Single-model evidence on public/A. (a) Discrete tested learning rates; dashed line: AdamW. (b) AUROC point estimates, not a significant ranking gain. Both use robust Platt.

2. Methodology

2.1. Prediction unit and leakage control

Each sample pairs a learning objective with a chronological, role-marked transcript; entity identifiers are excluded. Random row splits would leak repeated transcripts across folds.

Development calibration and bootstrap analyses group by `session_id`; artifacts retain the split protocols. Unlike longitudinal knowledge tracing [8], rows here can reuse a full transcript for different objectives.

Evaluation scope. Muon results are post-competition public/A evaluations, not submitted leaderboard entries. Each run froze predictions before scoring, but the broader learning-rate and variant search used earlier public/A feedback. Calibration coefficients were fixed from legacy development predictions. Bootstrap intervals condition on fixed predictions; they do not cover model selection or training randomness.

2.2. Encoder and context

We fine-tune Qwen3-4B-Instruct-2507 [7] for binary sequence classification with LoRA rank 256, alpha 512, dropout 0.05, SDPA attention, and one epoch (four GPUs, global batch 4).

Training uses ordinary right truncation; inference uses validated 35/65 Head-tail input, preserving the objective and allocating 10,000 content tokens within a 10,240-token maximum. The head returns two logits. Optional features are the last non-padding token’s 2,560-dimensional HF hidden-state index 28, not the final layer.

2.3. Muon inside LoRA

For a frozen base weight W , LoRA [1] learns $W' = W + sBA$ with low-rank factors B and A . AdamW rescales coordinates independently; Muon orthogonalizes matrix momentum via Newton-Schulz iterations [2]. Our archived implementation uses Nesterov momentum 0.95, five Newton-Schulz steps, and weight decay 0.1.

Figure 1 shows the resulting path. Muon operates on the two low-rank factors separately rather than on the materialized product (BA). After PEFT injection, the audit records **504 LoRA matrices plus one classification matrix on the Muon path, with no AdamW fallback**. Reload and inference smoke tests reproduce saved predictions.

The principal path uses one checkpoint, a linear head and fixed robust Platt calibration:

$$p' = \sigma(0.6027178495 \logit(p) + 0.1141993277).$$

The mapping was fitted jointly to legacy seed-42/3407 development predictions, grouping repeated sessions, and is reused without fitting to Muon test outputs. The public predictor exposes this mapping and Layer-28 export; its portable HF runtime still requires logit-parity checks against the original ms-swift template.

2.4. Optional nonlinear decoding

For comparison, a separately AdamW-adapted checkpoint supplies Layer-28 representations and native logits to XGBoost [3]. Five fixed configurations span depths 3–5, 372–2,749 trees, and learning rates 0.0068–0.0681; two use hidden states alone and three add native logits. Their probability outputs are averaged.

Because Log Loss penalizes confident mistakes disproportionately, we fit beta calibration

on session-grouped out-of-fold predictions:

$$\logit(p') = a \log(p) + b \log(1 - p) + c.$$

Beta calibration handles asymmetric probability error [4]. It is fitted on development predictions. The submitted pipeline also included a neural-turn branch and an objective prior; later tests found no stable prior gain.

In the clean session-grouped OOF calibration ablation, beta calibration reduces the five-model XGBoost ensemble’s Log Loss from **0.613955 to 0.589772** while preserving AUROC. This isolates a probability-calibration gain from unchanged ranking.

3. Extensions & Generalizability

3.1. Generalization beyond familiar identities

The training set contains 35,072 TSL examples. The released test set contains 10,508 examples across 8,241 sessions. Among test rows, **42.6% of students, 93.9% of questions, and 91.5% of learning objectives are unseen** relative to training. This novelty limits an identity-memorization explanation, but does not by itself establish dialogue reasoning.

Frozen XGBoost scores **0.596309 Log Loss and 0.647613 AUROC** overall. Unseen students reach 0.594261 and 0.659511; unseen questions and objectives are harder at approximately 0.602 Log Loss. A smoothed objective-frequency baseline achieves only 0.621481 Log Loss and 0.517346 AUROC, showing that objective frequency alone is insufficient.

The 868 Eedi examples come from an unseen source. A separate frozen audit compares AdamW robust Platt (0.703984 Log Loss / 0.554126 AUROC) with Muon beta (0.719158 / 0.564159). Higher ranking but worse Log Loss is not a uniform transfer gain; this is not the main robust-Platt comparison.

3.2. Head-tail is an input contract

A legacy AdamW fold-0 2×2 audit crossed ordinary and Head-tail training/inference formats on untouched development data, not public/A Muon predictions. For the ordinary-trained

model, Head-tail inference improved raw classification-head Log Loss by **0.019607** and frozen XGBoost-beta Log Loss by **0.002543**.

On that split, the raw Head-tail-trained classifier scores **0.544357** Log Loss on Head-tail input but **0.671186** on ordinary input, a Log Loss interaction of +0.107222. A separate frozen, Platt-calibrated turn-network audit on 7,015 untouched rows tests temporal information: using only the first half of turn pairs gives 0.564007 Log Loss, versus 0.541048 for the second half, a **0.022959** gain with session-bootstrap interval [**0.018048, 0.027754**]. Together these audits support preserving late evidence within a validated checkpoint-format pairing.

3.3. Objective-specific evidence

We identify **709 mixed-outcome sessions covering 1,625 rows**: the same transcript is paired with opposite labels across objectives. This motivates objective conditioning, not proof of correct evidence localization. Tutor praise is an error-analysis cue, never a post-processing rule.

3.4. What Muon changes

Muon requires optimizer-specific learning-rate tuning. Robust Platt Log Loss is **0.592714 at 2e-5, 0.591436 at 2.5e-5, 0.600297 at 3e-5, and 0.632914 at 1e-4**. Thus 2.5e-5 is our strongest standalone checkpoint (Figure 2). Against AdamW under the same robust Platt mapping, Log Loss improves by **0.002447**, with paired session-bootstrap interval [**0.000650, 0.004234**], and AUROC rises from **0.650021 to 0.653200**. Its gain interval [-0.001216, 0.007542] crosses zero, so the ranking improvement is a point estimate.

Only 5,028 of 10,508 rows improve, but corrections concentrate on costly errors: the decile defined by highest AdamW loss gains **0.064646** mean Log Loss and negative outcomes gain **0.028497**, while positive outcomes regress by 0.009474. Threshold changes are nearly balanced (185 AdamW-wrong to Muon-correct versus 174 in reverse), but mean probability also falls from 0.706620 to 0.699362. These descrip-

tive groups do not isolate a causal mechanism.

The effect is strongest on TSL data (+**0.003050**, interval [**0.001171, 0.004983**]) and mixed-outcome sessions (+**0.011354**, interval [**0.006358, 0.016260**]). The latter is a probability-error result, not evidence of stronger within-session rank discrimination.

Optional fusion. Using beta rather than robust Platt probabilities, a 0.4 AdamW Layer-28 XGBoost / 0.6 Muon head blend scores 0.590136/0.657522, versus 0.590509/0.657425 with the earlier 2e-5 head. Its Log Loss gain over the 2.5e-5 beta head is 0.001917, with session-bootstrap interval [0.000714, 0.003063]. This extension requires two checkpoints.

3.5. Controls and negative results

Muon adapters with an AdamW head yield 0.594456/0.647799 on public/A under robust Platt. The reverse control regresses untouched-development beta-calibrated Log Loss by 0.004508, with interval [**0.002084, 0.006976**]. Neither mixed-routing recipe improves its own reference; different splits and mappings preclude a matched factorial conclusion.

A second training epoch regresses performance. Label smoothing helps the raw classifier but not the calibrated downstream system. Checkpoint averaging, higher LoRA rank, smaller Qwen variants, and objective-prior post-processing all fail to produce stable final gains.

3.6. Conclusion and educational use

Muon-LoRA lowers observed Log Loss and raises the AUROC point estimate with one frozen-backbone model. For educational use, evaluate each objective separately, preserve late dialogue through a validated input contract, and audit calibrated probabilities as well as ranking. The output can prompt an independent item, explanation, or transcript review; it is not a causal teaching rating. Moderate AUROC and the Eedi shift rule out autonomous grading or high-stakes use.

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